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Quentin Rebergue - Software Engineer

Profile

Looking for a full-time position in software engineering with a focus on OS internals and adaptive memory management. MSc in Computer Engineering (EPITA - GISTRE) in Systems / Embedded. Fluent in French and English (TOEIC 915, C1). Interests: watchmaking, music, hiking/running, cooking.

Work Experience

- **Software Engineer - Platform team, ARISTA**, Dublin 02/2026 – Now

  Implemented non-zero reset value management for ASIC Physical Table registers across the chip-agnostic Strata infrastructure (SDK parsing, code generation, DMA) in python and C++. Won 1st place at Arista's Hackathon by building an AI agent optimization tool (70% token reduction on internal codebases). Approved for production development.

- **Head of Teaching Assistants - YAKA** / **ACU Program, EPITA**, Kremlin-Bicetre 09/2024 – 02/2026

  Led the national team of 67 TAs and supervised 5 intensive coding bootcamps (Java, C++, JS, C/Bash, SQL) and 7 large-scale projects (compiler, HTTP server in C, malloc, POSIX shell) for 700+ 3rd-year students across 5 campuses. Implemented 1-on-1 code review mentorship model, improving TA-to-student ratio from 1:30 to 1:20.

- **Freelance Software Engineer, Various Clients**, Paris 09/2021 – 03/2024

  Built and managed full-stack information systems for 10+ clients. Led B2B survey platform as first technical lead. Architected PIM + CRM + ERP ecosystem. Owned end-to-end IS.

School Projects

- **myMalloc** – Custom malloc implementation in C focusing on efficient memory reuse and fragmentation minimization with bucket-based allocation and free block tracking. Uses mmap for page management.
- **My Docker** – Built a lightweight container runtime in C, using Linux namespaces, pivot_root, capabilities dropping, seccomp syscall filtering, and cgroups. Pulls and mounts a root filesystem from a container registry.
- **RFID Driver** – Developed a Linux kernel character device driver for the MFRC522 RFID reader with file_operations callbacks, SPI regmap communication, and a FIFO command interface. Tested via QEMU emulation.
- **ARM project** – Built an STM32F429ZI embedded system acting as a UART command server using libgapcom (Protobuf over UART). Implemented a 4-level log subsystem and an MPU6050 gyroscope driver over I2C with FIFO reads.
- **Unicycle CPU** – Designed a 32-bit single-cycle ARM-like processor in VHDL (ALU, register file, instruction decoder, control unit, conditional branching), synthesized on a real FPGA with Intel Quartus, and validated with testbenches.

Education

- **MSc Computer Engineering, Low-level & Systems (GISTRE)**, EPITA 2020 – 2026
- **Preparatory Class (PTSI) in Engineering Science**, Lycée Jules Ferry Versailles 2019 – 2020

Technical Expertise

- **Programming & Languages:** ADA, ASM, C, C++, Python, Shell, VHDL.
- **Embedded & Hardware:** ASIC interfacing, I2C, SPI, UART, FPGA, RTL.
- **Systems:** DMA, Linux, cache coherence, drivers, kernel, multithreading, shared memory, virtualization.